$$i = \frac{\varepsilon}{R}$$

$$i = \frac{Blv}{R} \qquad (4.19)$$

Energy conservation

The current-carrying movable rod AB kept in the perpendicular magnetic field experiences a force $\vec{F}_B$ in the outward direction, opposite to its motion. This force is given by

$$\vec{F}_B = i\,\vec{l} \times \vec{B}$$

$$\left|\vec{F}_B\right| = ilB\sin\theta$$

$$F_B = ilB \qquad \text{since } \theta = 90°$$

In order to move the rod with a constant velocity $\vec{v}$, a constant force that is equal and opposite to the magnetic force, must be applied.

$$\left|\vec{F}_{app}\right| = \left|\vec{F}_B\right| = ilB$$

Therefore, mechanical work is done by the applied force to move the rod. The rate of doing work or power is

$$P = \vec{F}_{app} \cdot \vec{v} = F_{app}\, v \cos\theta \qquad \text{Here } \theta = 0°$$

$$= ilBv$$

$$= \left(\frac{Blv}{R}\right) lBv$$

$$P = \frac{B^2 l^2 v^2}{R}$$

When the induced current flows in the loop, Joule heating takes place. The rate at which thermal energy is dissipated in the loop or power dissipated is

$$P = i^2 R$$

$$P = \left(\frac{Blv}{R}\right)^2 R$$

$$P = \frac{B^2 l^2 v^2}{R} \qquad (4.21)$$

This equation is exactly same as the equation (4.20). Thus the mechanical energy needed to move the rod is converted into electrical energy which then appears as thermal energy in the loop. This energy conversion is consistent with the law of conservation of energy.

EXAMPLE 4.14

A circular metal of area 0.03 m^2 rotates in a uniform magnetic field of 0.4 T. The axis of rotation passes through the centre and perpendicular to its plane and is also parallel to the field. If the disc completes 20 revolutions in one second and the resistance of the disc is 4 Ω, calculate the induced emf between the axis and the rim and induced current flowing in the disc.

Solution

$A = 0.03$ m^2; $B = 0.4$ T; $f = 20$ rps;
$R = 4\ \Omega$

Area swept out by the disc in unit time
= Area of the disc × frequency

$$\frac{dA}{dt} = 0.03 \times 20$$

$$= 0.6 \text{ m}^2\text{s}^{-1}$$

The magnitude of the induced emf,

$$\varepsilon = \frac{d\Phi_B}{dt} = \frac{d(BA)}{dt} = B\frac{dA}{dt}$$

$$\varepsilon = \frac{0.4 \times 0.6}{1} = 0.24\,\text{V}$$

Induced current, $i = \frac{\varepsilon}{R} = \frac{0.24}{4} = 0.06\,\text{A}$

Note

Emf can be induced by changing relative orientation between the coil and the magnetic field. This can be achieved either by rotating a coil in a magnetic field or by rotating a magnetic field within a stationary coil. Here rotating coil type is considered.

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